10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 5 — Algebraic Fractions

Unit 5 | Review + MCQs | New Book (PECTAA)

Review Exercise 5 (Solved)

Chapter 5 is about algebraic fractions. In this review, first reduce each fraction to its lowest form, then simplify, and at the end solve the equations that reduce to quadratic form.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) The expression $(2x - 1)/(x^2 + 4)$ is:

(a) a polynomial
(b) an algebraic fraction
(c) a numerical fraction
(d) an equation

Answer: (b) an algebraic fraction — it is a ratio of two polynomials and the denominator has the variable x

(ii) Lowest form of $18a^2bc^2/12ac$ is:

(a) 9abc/6
(b) 3abc/2
(c) 18b/12
(d) 3/2

Answer: (b) 3abc/2 — divide both by 6ac: $18a^2bc^2 \div 12ac = 3abc/2$

(iii) Lowest form of 15xyz/10 is:

(a) 3xyz/2
(b) 3/2
(c) xyz
(d) xyz/2

Answer: (a) 3xyz/2 — divide both by 5: 15xyz/10 = 3xyz/2

(iv) Simplified form of (3x + 1)/5 + (2x + 3)/5 is:

(a) (5x + 4)/10
(b) (5x + 4)/5
(c) (5x + 4)/25
(d) 5x + 4

Answer: (b) (5x + 4)/5 — same denominator, so add the numerators and keep 5

(v) Simplified form of $[(x^2 - y^2)/(x + y)] \times (x - y)$ is:

(a) $(x + y)^2$
(b) $x^2 + y^2$
(c) $x - y$
(d) $(x - y)^2$

Answer: (d) $(x - y)^2$ — $(x + y)(x - y)/(x + y) = (x - y)$, then $(x - y)(x - y) = (x - y)^2$

(vi) $(x^3 - y^3) \div (x - y)$ in simplified form is:

(a) $x^2 - xy + y^2$
(b) $x^2 + xy + y^2$
(c) $x^2 - y^2$
(d) $x^2 + y^2$

Answer: (b) $x^2 + xy + y^2$ — $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

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(vii) An equation of the form $5/x + (1 - x)/3 = 1/(6x)$ is called:

(a) radical equation (b) reciprocal equation
(c) fractional equation (d) exponential equation

Answer: (c) fractional equation — the unknown x is in the denominator, so it is a fractional equation

(viii) An equation of the form $5^x + 64 \cdot 5^{-x} - 20 = 0$ is called:

(a) radical equation (b) exponential equation
(c) reciprocal equation (d) fractional equation

Answer: (b) exponential equation — the variable is in the power (index) of 5

(ix) Roots of $y^2 - 24y + 128 = 0$ are:

(a) 8, −16 (b) 8, 16
(c) −8, 16 (d) −8, −16

Answer: (b) 8, 16 — 8 + 16 = 24 and 8 × 16 = 128

(x) Linear factors of $3x^2 + 10x + 3 = 0$ are:

(a) (x + 3), (3x + 1) (b) (x + 3), (3x − 1)
(c) (x − 3), (3x + 1) (d) (x − 3), (3x − 1)

Answer: (a) (x + 3), (3x + 1) — $(x + 3)(3x + 1) = 3x^2 + 10x + 3$

2. Reduce the following to the lowest form.

(i)

$$\frac{x^2 - 7x + 12}{x^2 - 6x + 9}$$

$$= \frac{(x - 3)(x - 4)}{(x - 3)(x - 3)}$$

Cancel the common factor (x − 3)

Answer: = (x − 4)/(x − 3), x ≠ 3

(ii)

$$\frac{x^2 + 3x - 18}{7x - 21}$$

$$= \frac{(x + 6)(x - 3)}{7(x - 3)}$$

Cancel the common factor (x − 3)

Answer: = (x + 6)/7, x ≠ 3

3. Simplify.

(i)

$x^2 + x - 6 = (x + 3)(x - 2)$

$2x^2 - 7x + 6 = (2x - 3)(x - 2)$

$$= \frac{5}{(x + 3)(x - 2)} - \frac{1}{(2x - 3)(x - 2)}$$

L.C.M. = (x + 3)(x − 2)(2x − 3)

$$= \frac{5(2x - 3) - 1(x + 3)}{(x + 3)(x - 2)(2x - 3)}$$

$$= \frac{10x - 15 - x - 3}{(x + 3)(x - 2)(2x - 3)}$$

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$= \frac{9x - 18}{(x + 3)(x - 2)(2x - 3)}$

$= \frac{9(x - 2)}{(x + 3)(x - 2)(2x - 3)}$

Cancel the common factor $(x - 2)$

Answer: = 9/[(x + 3)(2x − 3)]

(ii)

$x^3 - 27 = (x - 3)(x^2 + 3x + 9)$

$x^2 - 9 = (x - 3)(x + 3)$

$x^2 + 6x + 9 = (x + 3)^2$

$= \frac{(x - 3)(x^2 + 3x + 9)}{(x - 3)(x + 3)} \times \frac{(x + 3)^2}{x^2 + 3x + 9}$

Cancel $(x - 3)$, $(x^2 + 3x + 9)$ and one $(x + 3)$

Answer: = x + 3

(iii)

$a^2 - (b - c)^2 = (a - b + c)(a + b - c)$

$(a + b)^2 - c^2 = (a + b - c)(a + b + c)$

$a^2 - (b + c)^2 = (a - b - c)(a + b + c)$

$(a - b)^2 - c^2 = (a - b - c)(a - b + c)$

$= \frac{(a - b + c)(a + b - c)}{(a + b - c)(a + b + c)} \times \frac{(a - b - c)(a + b + c)}{(a - b - c)(a - b + c)}$

Cancel all the common factors

Answer: = 1

4. Solve the following equations.

(i)

Put $y = x^2$, then x^4

$= y^2$

$y^2 - 16y + 63 = 0$

$y^2 - 7y - 9y + 63 = 0$

$y(y - 7) - 9(y - 7) = 0$

$(y - 7)(y - 9) = 0$

$y = 7 \quad \text{or} \quad y = 9$

Put back $y = x^2$

$x^2 = 7 \quad \text{or} \quad x^2 = 9$

Answer: x = ± √7, x = ± 3

(ii)

Put $y = x^2$, then x^4

$= y^2$

$4y^2 - 16y + 15 = 0$

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$4y^2 - 10y - 6y + 15 = 0$

$2y(2y - 5) - 3(2y - 5) = 0$

$(2y - 5)(2y - 3) = 0$

$y = 5/2$ or $y = 3/2$

Put back $y = x^2$

$x^2 = 5/2$ or $x^2 = 3/2$

Answer: $x = \pm \sqrt{10}/2$, $x = \pm \sqrt{6}/2$

(iii)

$3^{2x} - 12 \cdot 3^x + 27 = 0$

Put $y = 3^x$, then 3^{2x}

$= y^2$

$y^2 - 12y + 27 = 0$

$(y - 3)(y - 9) = 0$

$y = 3$ or $y = 9$

Put back $y = 3^x$

$3^x = 3^1$ or $3^x = 3^2$

Answer: $x = 1$, $x = 2$

(iv)

Put $y = x + \frac{1}{x}$

Then $x^2 + \frac{1}{x^2} = y^2 - 2$

$(y^2 - 2) - 3y - 2 = 0$

$y^2 - 3y - 4 = 0$

$(y - 4)(y + 1) = 0$

$y = 4$ or $y = -1$

Case $y = 4$

$x + \frac{1}{x} = 4$

$x^2 - 4x + 1 = 0$

$x = (4 \pm \sqrt{12})/2$

$= 2 \pm \sqrt{3}$

Case $y = -1$

$x + \frac{1}{x} = -1$

$x^2 + x + 1 = 0$

Discriminant $= 1 - 4$

$= -3 < 0$, so no real value

Answer: $x = 2 \pm \sqrt{3}$

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(v)

Put $y = x + \frac{1}{x}$

Then $x^2 + \frac{1}{x^2} = y^2 - 2$

$(y^2 - 2) + y = 4$

$y^2 + y - 6 = 0$

$(y + 3)(y - 2) = 0$

$y = -3 \quad \text{or} \quad y = 2$

Case $y = 2$

$x + \frac{1}{x} = 2$

$x^2 - 2x + 1 = 0$

$(x - 1)^2 = 0$

$x = 1$

Case $y = -3$

$x + \frac{1}{x} = -3$

$x^2 + 3x + 1 = 0$

$x = (-3 \pm \sqrt{5})/2$

Answer: $x = 1$, $x = (-3 \pm \sqrt{5})/2$

(vi)

$(x + 9)(x - 7) = x^2 + 2x - 63$

$(x - 3)(x + 5) = x^2 + 2x - 15$

Put $y = x^2 + 2x$

$(y - 63)(y - 15) = 385$

$y^2 - 78y + 945 = 385$

$y^2 - 78y + 560 = 0$

$y^2 - 70y - 8y + 560 = 0$

$y(y - 70) - 8(y - 70) = 0$

$(y - 70)(y - 8) = 0$

$y = 70 \quad \text{or} \quad y = 8$

Case $y = 70$

$x^2 + 2x - 70 = 0$

$x = (-2 \pm \sqrt{284})/2$

$= -1 \pm \sqrt{71}$

Case $y = 8$

$x^2 + 2x - 8 = 0$

$(x + 4)(x - 2) = 0$

$x = -4 \quad \text{or} \quad x = 2$

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Answer: $x = 2, x = -4, x = -1 \pm \sqrt{71}$

(vii)

$(x - 1)(x - 2) = x^2 - 3x + 2$

$(x + 5)(x - 8) = x^2 - 3x - 40$

Put $y = x^2 - 3x$

$(y + 2)(y - 40) = -360$

$y^2 - 38y - 80 = -360$

$y^2 - 38y + 280 = 0$

$y^2 - 28y - 10y + 280 = 0$

$y(y - 28) - 10(y - 28) = 0$

$(y - 28)(y - 10) = 0$

$y = 28 \quad \text{or} \quad y = 10$

Case $y = 28$

$x^2 - 3x - 28 = 0$

$(x - 7)(x + 4) = 0$

$x = 7 \quad \text{or} \quad x = -4$

Case $y = 10$

$x^2 - 3x - 10 = 0$

$(x - 5)(x + 2) = 0$

$x = 5 \quad \text{or} \quad x = -2$

Answer: $x = 5, x = -2, x = 7, x = -4$

5. Ahmad takes 2 hours to paint 50 glasses. Faiza takes 2 hours to paint 45 glasses. Working together, how long will it take them to paint 150 glasses?

Ahmad paints $\frac{50}{2} = 25$ glasses in 1 hour

Faiza paints $\frac{45}{2}$ glasses in 1 hour

Both together paint $25 + \frac{45}{2} = \frac{95}{2}$ glasses in 1 hour

Time for 150 glasses $= 150 \div \frac{95}{2}$

$= 150 \times \frac{2}{95} = \frac{300}{95}$

$= \frac{60}{19}$ hours

$= 3.16$ hours, that is about 3 hours 10 minutes

Answer: About 3 hours 10 minutes (60/19 hours)

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6. A tap can fill a tank in 6 hours. Another tap can fill it in 9 hours. If both taps are opened together, how long will it take to fill the empty tank?

First tap fills $\frac{1}{6}$ of the tank in 1 hour

Second tap fills $\frac{1}{9}$ of the tank in 1 hour

Together they fill $\frac{1}{6} + \frac{1}{9} = \frac{3+2}{18} = \frac{5}{18}$ in 1 hour

Time to fill the tank = $\frac{18}{5}$ hours

= 3.6 hours

= 3 hours 36 minutes

Answer: Both taps together fill the tank in 3 hours 36 minutes

7. Maham bought a certain number of toys for Rs. 300. If each toy had cost Rs. 5 less, she could have bought two more for the same amount. How many toys did she buy?

Let the price of one toy = Rs p and the number of toys = n

Then $p \times n = 300$

If each toy costs Rs 5 less, she gets 2 more toys

$(p - 5)(n + 2) = 300$

$pn - 5n + 2p - 10 = 300$

$300 - 5n + 2p - 10 = 300$

$2p = 5n + 10$

$p = (5n + 10)/2$

Put this value in $p \times n = 300$

$n(5n + 10)/2 = 300$

$5n^2 + 10n = 600$

$n^2 + 2n - 120 = 0$

$n^2 + 12n - 10n - 120 = 0$

$n(n + 12) - 10(n + 12) = 0$

$(n + 12)(n - 10) = 0$

$n = -12$ or $n = 10$

Number of toys cannot be negative, so n = 10 and p = Rs 30

Answer: Maham bought 10 toys at Rs 30 each

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